

Part 1

1. Explain the implications of 'align before fuse' on your tasks of interest. at which degree, and what type of fusion needs to be performed? And what level of alignment would you need to perform for your data, so that subsequent fusion or representation learning is successful?

For my **visuo-haptic material recognition** task, **align before fuse** implies that I should first learn **coordinated / aligned representations** across modalities before training a fusion module. This mirrors ALBEF's idea: establish **cross-modal coordination** (e.g., via contrastive objectives) so that subsequent fusion learns meaningful interactions rather than struggling with heterogeneous feature spaces.

Degree (stage) of fusion: I will use **Intermediate fusion** (a.k.a. mid-level fusion). Each modality is encoded separately into an embedding (a sequence encoder for haptics; an MLP for visual priors), and fusion happens at the embedding level rather than at raw input (**Early fusion**) or at the prediction level (**Late fusion**). This matches the course fusion taxonomy.

Type (mechanism) of fusion: As primary choices, I will compare **Additive fusion** (e.g., concatenation + MLP; a strong baseline) and **Gated fusion** (which explicitly learns to downweight unreliable modality signals). **Gated fusion** is especially relevant because my goal emphasizes cases where **vision is unreliable** (occlusion/clutter/ambiguous appearance).

Level/type of alignment needed: I primarily need **Explicit alignment** using **paired data** (haptic interaction $\leftrightarrow$ corresponding visual prior). Concretely, I will use **Contrastive alignment (InfoNCE-style)** so **positive pairs** are pulled together and **negative pairs** are pushed apart. This is the course's "explicit alignment via contrastive learning" framing and is conceptually consistent with ALBEF's contrastive alignment stage. I do **not** expect to need **Continuous alignment** (e.g., DTW-style warping) because my visual modality is a static prior vector rather than a continuous sequence; the alignment objective is naturally **discrete paired alignment** at the sample/interaction level.

2. How can you perform controlled experiments to validate the types of fusion and/or alignment needed for your tasks? What are some challenges you foresee in fusing and aligning your data?

To validate the necessary **degree/type of fusion** and **level/type of alignment**, I will run **controlled ablations** along three axes:

- **Fusion degree/stage ablation: Early vs Intermediate vs Late fusion** (keeping unimodal encoders comparable).
- **Fusion type/mechanism ablation: Additive fusion vs Gated fusion** (optionally a **Multiplicative** variant if needed).
- **Alignment ablation:** no explicit alignment vs **Explicit contrastive alignment (InfoNCE)** vs (optionally) **CCA-style alignment** as a classical baseline.

To make these experiments task-relevant, I will introduce **controlled reliability shifts**: I will perturb the visual priors to simulate unreliable vision (e.g., increase entropy toward uniform, inject noise, or apply dropout), and evaluate how different fusion types behave under degradation. **Gated fusion** should be measurably more robust if it truly learns to suppress corrupted visual signals.

Challenges I foresee:

- **Heterogeneity:** haptic data are **time-series** with rich dynamics, while visual priors are static vectors, making representation coordination nontrivial.
- **Shared vs unique information:** haptics may contain **unique physical information** (e.g., thermal/vibration cues) not present in vision; over-emphasizing shared information could hurt performance.
- **Class imbalance/noise:** imbalance can bias both classification and contrastive negatives; noisy priors can create spurious alignment signals.
- **Redundancy assumptions may fail:** “multi-view redundancy” assumptions often do not directly transfer to multimodal settings.

3. Explain the implications of 'platonic representation hypothesis' on your tasks of interest. Do you believe alignment between modalities would automatically emerge as models trained on your data are scaled up?

The **Platonic Representation Hypothesis (PRH)** suggests that as models improve and scale, representations learned from different modalities may **converge** toward a shared underlying

structure (a “statistical model of reality”), which in the multimodal setting appears as **Implicit (Emergent) Alignment**.

For my task, PRH would imply that sufficiently strong unimodal haptic and unimodal visual models might independently learn compatible “material structure” embeddings (e.g., capturing latent physical properties), making fusion easier.

However, I do **not** expect alignment to **automatically emerge** strongly in my specific pipeline simply by scaling. My visual modality is not raw images but **derived material probability vectors**, which may bake in upstream biases, and CLAMP-scale data is far smaller than the regimes where PRH evidence is strongest. Therefore, I expect **Explicit alignment** (e.g., **contrastive learning**) will still be necessary to reliably support downstream fusion.

4. What are some reasons why alignment would not emerge i.e., counter-arguments to the platonic representation hypothesis? You are encouraged to search for follow-up works to the original paper that both support and counter the original arguments.

A) In general: why alignment might *not* emerge

1. Modality-unique information (shared vs. unique factors):

Even if two modalities observe the same “world,” each modality can contain **unique information** that is not recoverable from the other. If the optimal solution keeps those unique factors, models may not converge to a single shared representation; at best, they may share only partial structure.

2. Shortcut learning / task sufficiency:

Models can solve downstream tasks using **spurious correlations** or modality-specific shortcuts without learning the underlying shared latent structure. In that case, scaling can improve task performance without producing strong cross-modal alignment.

3. Objective mismatch:

PRH-style emergence is more plausible when training objectives force models to capture broadly useful structure. If the objective is narrow (single task, limited diversity), there is

less pressure for representations to converge across modalities.

4. **Measurement ambiguity (alignment $\neq$ behavioral equivalence):**

“Alignment” depends heavily on *how it is measured*. Different representational similarity metrics can disagree, and similarity in a metric does not necessarily imply similar behavior. A broad framework emphasizing this distinction is discussed in “**Getting Aligned on Representational Alignment**”.

B) In my setting (CLAMP visuo-haptic material recognition): why emergence is even less likely

1. **My visual modality is not raw vision:**

I am using **image-derived material probability vectors** (a compressed, model-dependent representation), not raw images. This can bake in upstream biases and remove information that might otherwise support emergent alignment.

2. **Haptics has strong modality-unique signals:**

Haptics includes **thermal dynamics** and **micro-vibration** cues that may not have stable correlates in a visual prior vector. This increases the chance that optimal representations retain **unique haptic factors** rather than converging to a shared space.

3. **Limited scale relative to PRH regimes:**

Emergent alignment is most plausible at large scale and broad supervision. In CLAMP-scale settings, it is more realistic that I need **Explicit alignment** (e.g., contrastive learning) to reliably align modalities.

C) Follow-up / related works (support vs. counter/qualify)

Works that *challenge/qualify* PRH-style claims (often by critiquing metrics/methodology):

- “**Revisiting the Platonic Representation Hypothesis: An Aristotelian View**” (Fabian Gröger, Shuo Wen, Maria Brbić, 2026): argues commonly used similarity metrics can be **confounded by scale** and shows that some apparent convergence can **shrink after calibration**, with a more nuanced conclusion that **local neighborhood** agreement may persist.
- “**Getting Aligned on Representational Alignment**” (Ilia Sucholutsky et al., 2023; published in TMLR 2025): emphasizes that representational similarity depends on definitions/metrics and that alignment does not automatically imply **behavioral equivalence**.

Works that *support* PRH-adjacent convergence/compatibility intuitions (e.g., functional compatibility across models):

- **Model stitching / compatibility line of work:** evidence that independently trained networks can sometimes be “stitched” with limited loss is often used as **functional** evidence of representation compatibility (supportive of convergence intuitions).

At the same time, stitching as a similarity measure is actively debated:

- “**How Not to Stitch Representations to Measure Similarity: Task Loss Matching versus Direct Matching**” (András Balogh, Márk Jelasity, 2024/2025) argues common stitching procedures can be **misleading** as a similarity index.

5. What experiments would you propose to validate the existence and emergence of alignment in your tasks?

The key question is whether representations from different modalities naturally converge toward a shared structure as models scale, even without explicitly enforcing cross-modal alignment objectives.

To evaluate this in my visuo-haptic material recognition task, I would design experiments that test both the **existence of alignment at a fixed scale** and the **emergence of alignment as models scale up**, following experimental ideas similar to those used in the PRH paper.

A. Existence tests (alignment at a fixed scale)

First, I would test whether cross-modal representations are already aligned at the current model scale.

Cross-modal retrieval.

Given a **haptic embedding**, retrieve the corresponding **visual-prior embedding** from a batch (and vice versa). High retrieval accuracy indicates that the two modalities share a compatible embedding structure.

Neighborhood consistency (PRH-style metric).

The PRH paper evaluates alignment by comparing **nearest-neighbor structures** across representation spaces. Similarly, I would measure whether samples that are neighbors in the **haptic embedding space** also tend to be neighbors in the **visual embedding space**.

Consistent neighborhood structures suggest shared latent organization.

Cross-modal transfer probes.

Train a classifier on **haptic embeddings** and test it on **visual embeddings** (and vice versa). If the classifier generalizes across modalities, it indicates that the modalities encode **similar semantic structure**, which is another signal of alignment.

B. Emergence tests (alignment under scaling)

To test whether alignment **emerges naturally**, I would analyze how alignment metrics change as model scale increases.

Scaling experiments.

Train multiple models with increasing **model capacity** and/or **dataset size**, and track alignment metrics such as:

- cross-modal retrieval accuracy
- neighborhood consistency
- representation similarity metrics

If **implicit alignment emerges**, these metrics should improve as the model scales.

Training objective comparison.

Following the PRH hypothesis, I would compare two training regimes:

1. **No explicit alignment objective**
(only supervised classification + fusion)
2. **Explicit alignment objective**
(e.g., **contrastive alignment** using InfoNCE)

If PRH-style emergence holds for this task, the model trained **without explicit alignment** should still gradually develop **Implicit Alignment** as scale increases, reducing the performance gap with the explicitly aligned model.

6. Can you also think of some downsides of strongly (or perfectly) aligning your data modalities? How can you design experiments to validate that these risks are not present in your trained models?

Strong (or “perfect”) alignment can be harmful in multimodal settings:

- **Loss of unique information:** forcing haptic embeddings to match vision-based embeddings may discard modality-specific physical cues (course concept: **shared vs unique**, and redundancy assumptions may fail).
- **Bias propagation:** if visual priors contain systematic bias, strong alignment can inject that bias into haptic representations.
- **Brittleness under missing/corrupted modalities:** over-coupled representations may fail when vision is unreliable—precisely the scenario my project targets.

Experiments to validate risks are not present:

- **Unimodal integrity test:** evaluate haptic-only performance before vs after applying strong cross-modal alignment; large drops indicate loss of haptic-unique information.
- **Missing-modality / corruption robustness:** remove or corrupt visual priors at test time and check whether the model remains accurate and well-calibrated; robust fusion should degrade gracefully, and **Gated fusion** should help.
- **Shared-vs-unique diagnostic:** compare models that optimize only shared-information alignment vs models that preserve unique factors (inspired by the course discussion of shared+unique representations).

Part 2

Problem 3

1. (5 points) Try to get the best performance out of each model by playing around with hyperparameters (hint: you may have to playing around and even add additional arguments to the layers like dropout, look at the documentation and look into how we can improve performace). List the best performance you were able to get and the hyperparameters you used.

I performed iterative single-variable hyperparameter tuning for both unimodal models (**audio-only** and **image-only**). The best configuration for each model was selected based on **final test accuracy**.

Best Audio Model

Best hyperparameters:

- Learning rate: **1e-3**
- Weight decay: **1e-3**
- Epochs: **10**
- Batch size: **128**
- Dropout: **0.0**

Performance:

- Best validation accuracy: **41.68%**
- Best test accuracy: **41.17%**

Best Image Model

Best hyperparameters:

- Learning rate: **1e-3**
- Weight decay: **1e-3**

- Epochs: **6**
- Batch size: **128**
- Dropout: **0.3**

Performance:

- Best validation accuracy: **68.92%**
- Best test accuracy: **64.57%**

Observations from Hyperparameter Tuning

- **Audio model:**
Adding dropout (**0.2**) reduced test performance, and lowering the learning rate (**5e-4**) also resulted in lower test accuracy.
- **Image model:**
Moderate-to-strong dropout (**0.3**) improved generalization compared with **0.1** and **0.2**.
- **Epochs:**
Increasing the number of epochs did not always improve test performance. In some cases, validation accuracy increased while **test accuracy decreased**, indicating potential **overfitting**.

2. (5 points) Compare the performances of each modality. What do these suggest to you? What could be done to get the worst performing ones to get closer to the best performing modality/model?

The image modality significantly outperformed the audio modality:

- **Image test accuracy:** 64.57%

- **Audio test accuracy:** 41.17%

This suggests that, for **AV-MNIST in this setup**, image inputs provide **much stronger discriminative information** than audio inputs. The audio branch is likely limited by either **noisier signal quality**, **weaker feature extraction capacity**, or **less separable class structure** in the current representation.

What This Suggests

- **Not all modalities are equally informative in practice.**
- Stronger unimodal performance in one modality (**image**) does not automatically transfer to weaker modalities (**audio**).
- Improvements on the validation set for the weaker modality do not always translate to improvements on the test set, suggesting a **higher risk of overfitting**.

How to Improve the Worse Modality (Audio)

1. Improve audio encoder capacity

- Add deeper convolutional blocks
- Use batch normalization or residual connections
- Introduce attention pooling

2. Better audio preprocessing or feature extraction

- Normalize log-mel spectrograms
- Apply SpecAugment
- Use denoising or time–frequency masking

3. Regularization and optimization tuning

- Learning-rate scheduling
- Early stopping

- Label smoothing
- Gradient clipping

4. Handling noise or class imbalance

- Weighted loss functions
- Robust training strategies if audio features are noisier

5. Knowledge transfer from the stronger modality

- Distillation from an **image model** or **multimodal teacher** to an **audio student**

6. Multimodal training strategies

- Use multimodal fusion with **auxiliary unimodal losses** so the weaker modality receives stronger supervised gradients.

In summary, the results reveal a **clear modality gap (image $\gg$ audio)**. Closing this gap likely requires **targeted improvements in audio representation learning**, rather than only generic hyperparameter tuning.

Problem 4

1. (2 points) Sometimes when training you may notice the model gets stuck in a range of loss and never seems to get it's loss down. What does this suggest? What are some ways you can fix this?

When the training loss plateaus in a narrow range and does not decrease further, it typically indicates an **optimization plateau** or **representation mismatch**. The model may be stuck in a region of the loss landscape where gradients provide little improvement. In multimodal learning,

this can also occur when **heterogeneous modalities are not well aligned**, forcing the model to simultaneously learn alignment and fusion.

From the **Align Before Fuse (ALBEF)** perspective, this suggests that unimodal features may lie in different representation spaces, making cross-modal interactions difficult to learn directly.

Possible fixes include:

- **Introduce an alignment objective** before fusion (e.g., **contrastive alignment** using ITC / InfoNCE).
- **Improve optimization** (learning rate tuning, warmup + decay schedules, gradient clipping, proper weight decay).
- **Normalize features** (BatchNorm / LayerNorm) to prevent one modality from dominating.
- **Improve negative sampling** (e.g., harder negatives) to strengthen cross-modal discrimination.
- **Reduce supervision noise**, for example through teacher-based soft targets or momentum distillation.
- **Adjust model complexity**, starting from strong baselines (e.g., concatenation / additive fusion) before adding more expressive fusion modules.

2. (2 points) What are some other fusion methods we could use that we could use? Would they lead to improvements compared to early fusion?

Besides the current **feature-level concatenation**, other fusion strategies include:

1. **Output-level late fusion (decision ensemble)**: train unimodal classifiers separately and combine logits or probabilities.
2. **Multiplicative / bilinear fusion**: explicitly model pairwise cross-modal interactions.
3. **Tensor fusion**: capture higher-order interactions across modalities.

4. **Low-Rank Multimodal Fusion (LMF)**: a computationally efficient approximation of tensor fusion.
5. **Gated / dynamic fusion**: learn sample-wise modality weights depending on modality reliability.
6. **Cross-attention / Transformer fusion**: learn fine-grained cross-modal dependencies between representations.

Would these always improve over simple **concatenation (early fusion)**? **Not necessarily.**

More expressive fusion methods can help when **cross-modal interactions are strong and informative**. However, with **limited data, noisy labels, or weak cross-modal dependency**, complex fusion modules may **overfit** and fail to outperform simple baselines like concatenation.

3. (6 points) Explain the difference between early fusion techniques and late fusion techniques.

The key difference between **early fusion** and **late fusion** is **where in the model pipeline the modalities are combined**.

Early fusion combines modalities at the **input or low-level feature stage**, allowing the model to learn joint representations from the beginning.

Late fusion first learns **modality-specific representations** using separate encoders and then combines high-level features or predictions.

Early fusion – benefits

- Enables **joint modeling** early in the network.
- Can capture **fine-grained cross-modal interactions** when modalities are well aligned.

Early fusion – trade-offs

- Sensitive to **alignment errors** and modality heterogeneity.
- Harder to optimize if modalities have different scales or distributions.
- Less robust when a modality is **missing or noisy**.

Late fusion – benefits

- More **modular and stable** for heterogeneous modalities.
- Allows each modality to learn **strong unimodal representations** first.
- Often more robust when one modality is **weak, noisy, or missing**.

Late fusion – trade-offs

- May miss **low-level interactions** (e.g., token-level or frame-level interactions).
- If the task requires strong early cross-modal coupling, performance may be limited.

A practical strategy, consistent with lecture concepts and **ALBEF**, is to **align modalities first and then fuse them**, which improves optimization stability and allows the fusion module to focus on meaningful cross-modal interactions.

Problem 5

**** (5 points) **** In your write up, report the best validation accuracies of your multimodal model (don't forget to include what hyperparameters you included) after training and any modifications that had to be done to your data or model to be able to train on it. In addition, talk about which technique you believe would be best for your dataset and why that is.

In Problem 5, I implemented and compared four multimodal fusion methods on my **CLAMP dataset**:

- Early Fusion
- Late Fusion
- Tensor Fusion
- Low-Rank Tensor Fusion (LMF)

The dataset contains two modalities:

1. **Haptic force-related time series** (Force, Force diff, Impedance)
2. **Visual prior probabilities** (gpt_probs) mapped to **14 material classes**

Training Setup and Hyperparameters

- **Optimizer:** AdamW
- **Learning rate:** 1e-3
- **Weight decay:** 1e-4
- **Epochs:** 15
- **Batch size:** 128
- **Loss function:** CrossEntropyLoss

Embedding dimension:

- 64 for most models
- 48 for Tensor Fusion (to control parameter growth)

LMF rank: 8

Dataset split:

- Random **train/val/test split = 80 / 10 / 10**
- Fixed random seed for reproducibility

Data and Model Modifications

To ensure stable multimodal training, several preprocessing and architectural adjustments were applied.

1. Modality-specific preprocessing

- **Haptic channels:** globally standardized using mean/std normalization
- **Visual prior vectors:** normalized to ensure they form valid probability distributions

2. Modality-specific encoders

- **1D CNN encoder** for haptic time-series signals
- **MLP encoder** for visual prior vectors

3. Tensor Fusion architecture adjustment

Tensor Fusion has very large parameter growth due to outer-product interactions, so the **embedding dimension was reduced to 48** to keep the model computationally manageable.

4. Consistent training controls

To ensure fair comparison across fusion methods:

- Fixed **random seed**
- Identical **optimizer and loss function**
- Same **training split**

Best Validation Performance

Model	Validation Accuracy
Early Fusion	84.19%
Late Fusion	84.27%
Tensor Fusion	85.43%
LMF Fusion	85.04%

Reference unimodal baselines:

- Visual prior model: **84.43%**
- Haptic-only model: **33.69%**

Best Technique for This Dataset

For this dataset, **Tensor Fusion** appears to be the best-performing technique because it achieved:

- **Highest validation accuracy:** 85.43%
- **Highest test accuracy among multimodal models:** 83.96%

A likely reason is that **material recognition benefits from higher-order cross-modal interactions between haptic dynamics and visual priors**. Tensor Fusion explicitly models these interactions through **outer-product style feature combinations**, allowing richer cross-modal representations.

However, **LMF (Low-Rank Multimodal Fusion)** is also a strong practical alternative. Its performance is very close to Tensor Fusion while using **significantly fewer parameters**, making it more suitable for **resource-constrained settings**.

Overall conclusion:

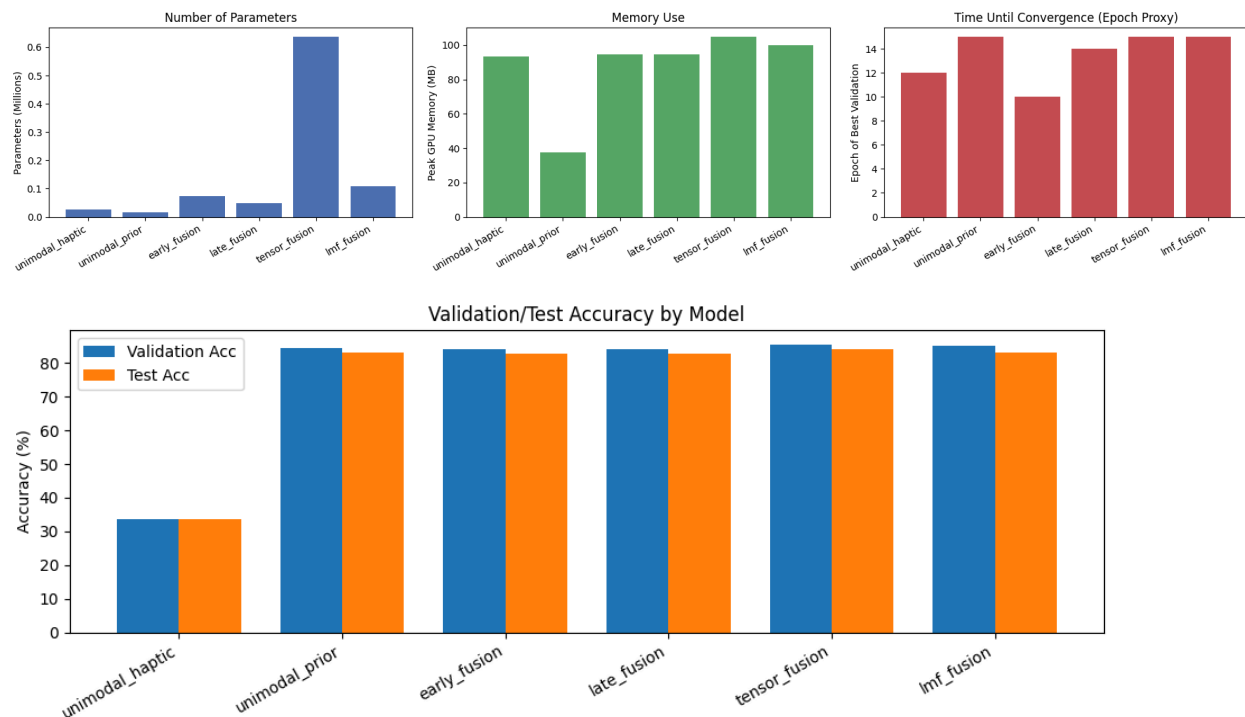
Multimodal fusion improves performance over the haptic-only model, and **higher-order interaction models (Tensor Fusion / LMF)** are particularly beneficial for tasks where modalities provide **complementary physical and semantic information**, such as material classification.

(10 points) In addition, create some visualizations of the following for each fusion:

- * Number of parameters for each model (unimodal and multimodal)
- * Memory Use
- * Time until convergence

You are free to plot them here or through other means (like wandb). After doing so, discuss what are the pros and cons of unimodal versus multimodal models.

The visualizations are shown below:



Pros and Cons of Unimodal vs. Multimodal Models

Unimodal Models

Pros

- **Computational efficiency:** Unimodal models are generally simpler and require fewer parameters, making them faster to train and less memory-intensive.

- **Implementation simplicity:** They are easier to implement, tune, and debug compared to multimodal architectures.
- **Strong baseline performance:** In my results, **unimodal_prior** achieved strong performance (~84% validation, ~83% test accuracy) while having the **smallest parameter count and lowest memory usage** among the strong models.

Cons

- **Limited information:** Unimodal models cannot capture complementary information across different modalities.
- **Performance depends heavily on modality quality:** If the chosen modality is weak or noisy, performance can degrade significantly. For example, **unimodal_haptic** performed much worse (~34% validation/test accuracy), demonstrating the limitations of relying on a single weak modality.

Multimodal Models

Pros

- **Leverage complementary signals:** Multimodal models combine information from multiple modalities, which can improve robustness and predictive performance.
- **Better performance when modalities are complementary:** In my experiments, multimodal fusion methods significantly outperform the **haptic-only model**, and the best multimodal model (**tensor_fusion**) achieved the highest accuracy (~85.4% validation, ~84.0% test).

Cons

- **Higher computational cost:** Multimodal models typically require more parameters, more GPU memory, and longer training times.
- **Higher implementation complexity:** Designing and tuning multimodal architectures (especially advanced fusion methods like Tensor Fusion) can be more difficult. For example, **tensor_fusion** achieved the best accuracy but also had the **largest model size and highest memory cost**.

Practical Takeaway for This Dataset

- If the goal is **maximum accuracy**, **tensor_fusion** is the best choice.
- If the goal is **efficiency with strong performance**, simpler methods such as **early_fusion**, **late_fusion**, or even **unimodal_prior** are competitive.

Overall, **multimodal models can provide accuracy improvements**, but these gains must be balanced against **increased computational cost and training complexity**.

Problem 6

1. (5 points) Any suprising results from using this on your dataset?

Yes—there were several surprising observations when applying this method to my dataset.

1. Meaningful cross-modal alignment despite low retrieval metrics

Although the overall retrieval metrics were modest (**Recall@1 = 3.50%**, **Recall@5 = 9.50%**), the model still learned some meaningful cross-modal structure. The similarity heatmap was not purely random, and sample-level inspection revealed cases where the correct pair was ranked first. This was surprising because it suggests that cross-modal alignment can still emerge even between **highly heterogeneous modalities**, such as haptic time series and visual-prior vectors.

2. Some failure cases were semantically close to correct

In several failed examples, the retrieved top-1 visual-prior distribution still had **similar dominant classes** to the ground-truth paired prior. This indicates that the model may capture **coarse semantic correspondence** even when strict instance-level matching fails.

3. Optimization improvements did not strongly improve retrieval

Increasing training to **100 epochs** significantly reduced the training loss (**2.6487**), but this did not lead to a proportional improvement in retrieval accuracy. This was unexpected and suggests that **lower loss does not necessarily translate to better instance-level retrieval performance** in this setting.

Overall Takeaway

These results suggest two main insights:

1. **Cross-modal alignment can still emerge even under strong modality mismatch.**
2. The learned alignment in this model appears to be **stronger at the semantic or distributional level** than at **exact one-to-one instance matching**.

2. (5 points) Typically, cross-entropy loss is used in this contrastive learning, why is this the case?

Cross-entropy loss is commonly used in contrastive learning because the problem can be formulated as a **classification task over pairwise matches**.

For each query embedding, the model must **identify its correct paired sample among a set of candidate samples in the batch**. Given a similarity matrix S (in this case computed using einsum), the **diagonal elements represent positive pairs**, while the **off-diagonal elements represent negative pairs**.

Using cross-entropy with a **softmax over similarity scores** allows the model to treat the correct pair as the target class and all other pairs as competing negatives.

This formulation is effective for several reasons:

- **Simultaneous attraction and repulsion:**
It pulls positive pairs closer together while pushing negative pairs apart.
- **Efficient use of in-batch negatives:**
Every other sample in the batch naturally acts as a negative example, which improves training efficiency.
- **Stable optimization:**
Softmax normalization provides well-behaved gradients and stable training dynamics.

- **Alignment with InfoNCE objective:**
Cross-entropy directly implements the **InfoNCE-style contrastive objective**, often applied symmetrically across rows and columns of the similarity matrix.

Overall, cross-entropy provides a **simple, stable, and computationally efficient way to learn discriminative cross-modal embeddings**.

3. (10 points) Create some visual examples of the data post alignment. Can you point out samples where the alignment worked and where it failed? Why do you suspect that is?

To analyze the behavior of the model after alignment, I visualized cross-modal similarity at both **global** and **sample** levels.

The visualizations include (figures are attached below):

- **Figure 1:** Global cross-modal similarity heatmap (haptic query embeddings vs. visual-prior candidate embeddings).
- **Figure 2:** Top-k similarity comparison for one **successful** and one **failed** query.
- **Figures 3–6:** Sample-level alignment examples (two successes and two failures).

For the sample-level analysis (Figures 3–6), each example is shown using a **three-panel visualization**:

- **Left panel:** Query **haptic time-series signal** (Force, Force diff, Impedance) over time.

- **Middle panel:** Retrieved **top-1 visual-prior distribution** (model prediction).
- **Right panel:** **Ground-truth paired visual-prior distribution**.

Note that in this dataset the “visual” modality does **not** correspond to raw image pixels. Instead, it is a **14-dimensional probability vector (gpt_probs)** representing **visual priors over material classes**. Therefore, the visualizations illustrate alignment between **haptic dynamics** and **visual prior distributions**, rather than haptic-to-raw-image alignment.

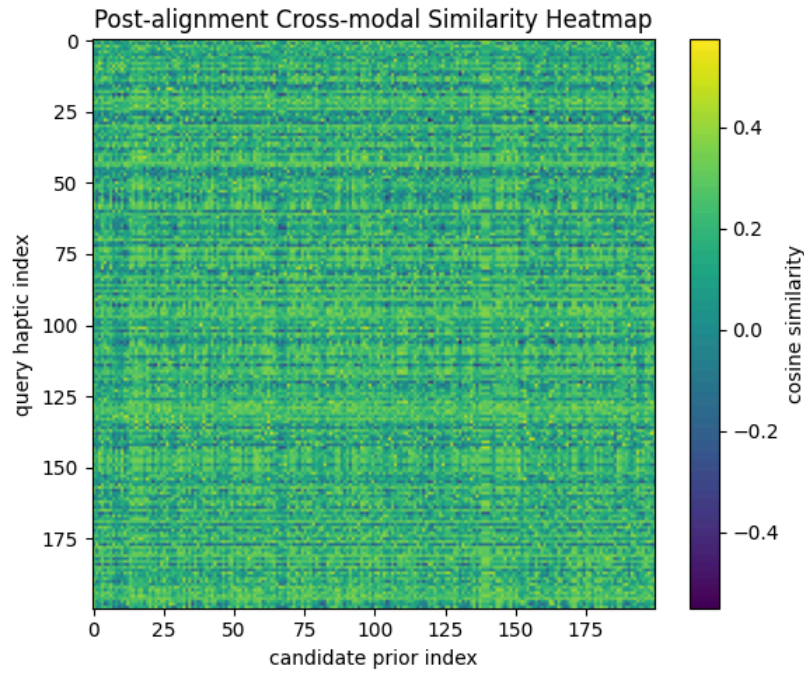


Figure 1

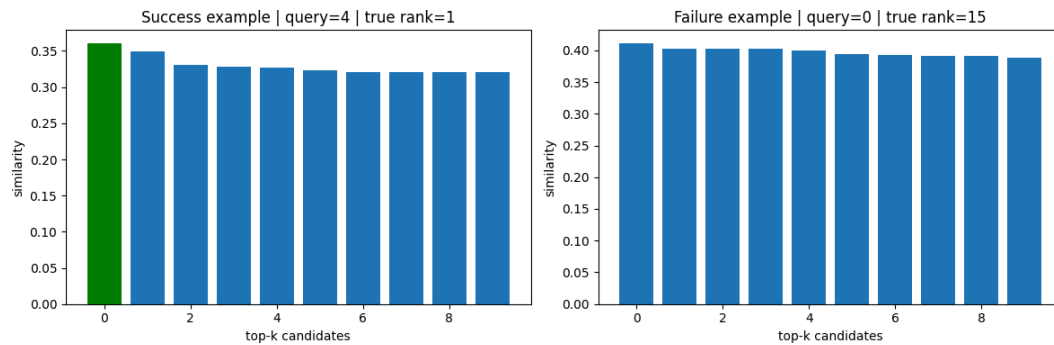


Figure 2

SUCCESS case | true_rank=1 (smaller is better)

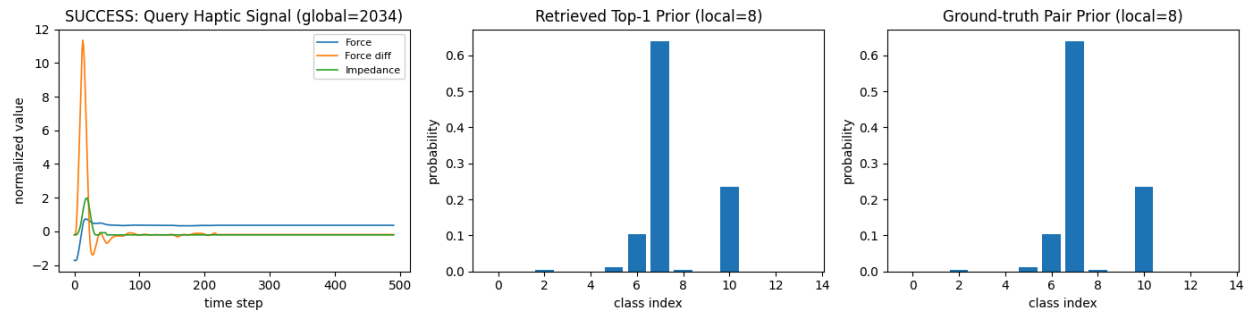


Figure 3

SUCCESS case | true_rank=1 (smaller is better)

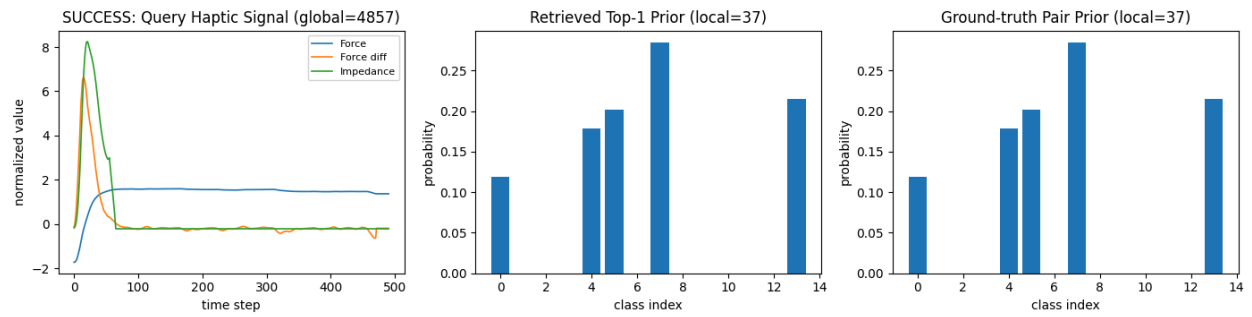


Figure 4

FAILURE case | true_rank=11 (smaller is better)

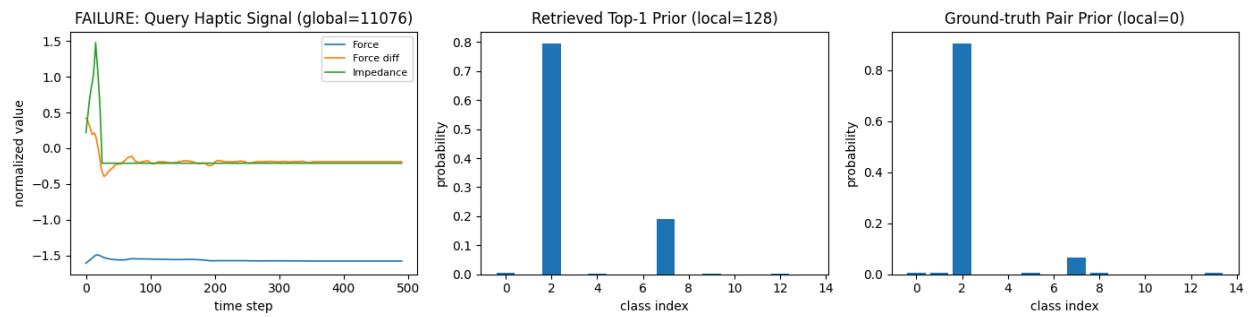


Figure 5

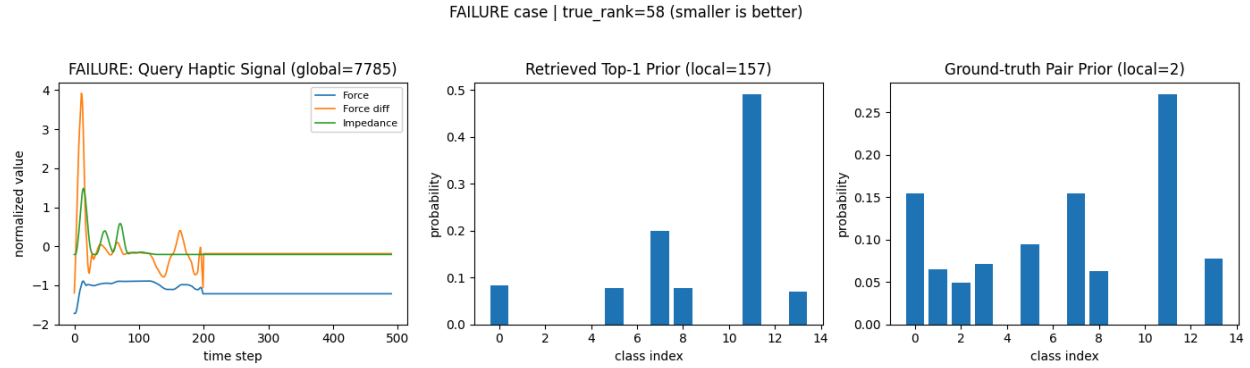


Figure 6

Global Alignment Behavior

At the global level, the post-alignment similarity matrix (**Figure 1**) does **not exhibit a strong diagonal structure**.

This observation is consistent with the retrieval metrics:

- **Recall@1 = 3.50%**
- **Recall@5 = 9.50%**

These results indicate that **exact one-to-one instance matching remains difficult**, even after training.

However, the heatmap still contains **structured regions**, suggesting that the model has learned **coarse cross-modal alignment**, even though precise pairwise alignment is weak.

Ranking-Level Observations

Figure 2 compares the **top-k similarity scores** for one successful retrieval and one failure case.

An interesting observation is that **top candidates often have very similar similarity scores**:

- In the **successful example**, the correct match ranks first but only by a **small margin**.
- In the **failure example**, the top candidates have **nearly identical similarity values**.

This suggests that the learned embedding space contains **high ambiguity**, where multiple candidates appear equally plausible.

Successful Alignment Examples

Figures **3 and 4** show examples where the alignment **worked correctly** (**true_rank = 1**).

In both cases:

- The retrieved **top-1 visual-prior distribution** closely matches the **ground-truth prior distribution**.
- The query haptic signals contain **distinct transient patterns**, which likely provide informative cues for correct retrieval.

These examples demonstrate that the model can successfully learn meaningful cross-modal correspondences when the signals are distinctive.

Failed Alignment Examples

Figures **5 and 6** show examples where the alignment **failed** (**true_rank = 11** and **true_rank = 58**).

Interestingly, these failures are **not always semantically incorrect**:

- The retrieved and ground-truth prior vectors often share **similar dominant classes**.
- The main difference lies in **fine-grained probability distributions**, which affects ranking.

This suggests a **many-to-many mapping problem**, where multiple candidate samples share similar prior distributions and therefore appear similarly compatible with the same haptic query.

Why Alignment Fails in Some Cases

There are several likely reasons for these failures:

1. **Modality mismatch**

Haptic signals are **continuous time-series dynamics**, while visual priors are **compressed semantic probability vectors**, making precise alignment difficult.

2. **Material property overlap**

Many materials share similar tactile characteristics, which increases retrieval ambiguity.

3. **Embedding ambiguity**

The similarity scores between top candidates are often very close, making the ranking unstable.

Overall Conclusion

These visualizations suggest that the model has learned **meaningful cross-modal alignment at a semantic or coarse level**, but **precise instance-level matching remains weak**.

The primary challenge appears to be **retrieval ambiguity caused by modality mismatch and overlapping material semantics**, rather than a complete failure of cross-modal representation learning.

Problem 7

1. (5 points) What concept did you find the most interesting?

The most interesting concept for me was the interaction between fusion and alignment, especially the idea that strong multimodal performance is not only about combining modalities, but about how well they are aligned before or during fusion. In my experiments, Tensor Fusion and LMF improved validation accuracy over simple baselines in supervised classification, but contrastive learning showed that lowering training loss does not automatically guarantee strong one-to-one retrieval alignment. This made the lecture theme (“align before fuse”) very concrete: fusion can exploit complementary information, but alignment quality determines whether that information is actually usable at the instance level.

2. (5 points) Which concepts (if any) do you see being useful towards your goal? Why? If there was none, discuss why.

Several concepts are directly useful for my goal (multimodal material recognition with haptic signals and visual priors).

First, heterogeneity-aware fusion design (early/late/tensor/LMF) is essential because my modalities have very different structures (time series vs. probability vectors).

Second, einsum-based interaction modeling is very useful for implementing and understanding multiplicative and tensor-style cross-modal interactions in a transparent way.

Third, contrastive alignment is valuable for learning a shared space across modalities, even when strict retrieval is difficult; it provides a complementary objective to supervised classification.

Finally, the emphasis on trade-offs (accuracy vs. parameter/memory cost) is highly practical: full tensor fusion gave the best accuracy, while LMF offered a better efficiency-performance balance, which is important for scalable deployment.